

Problem 3:

Rock Bed	Rock Characteristics	thickness, m	RQD, %	Average length of core, cm	σ_c , kg/cm ²	Cavability Index, 'I'
1	CgMg sst	4.4	98	24	210.68	4005
2	Intercalation	0.4	79	17	411.13	1174
3	CgFg sst	3.8	93	22	269.17	4284
4	Intercalation	0.76	97	23	411.13	3087
5	CgFg sst	12	87	20	293.91	7414
6	coal	0.67				0
7	laminated bed of shale and Intercalation	3.78	80	18	362.29	3386
8	Sandstone	2.4	97	23	439.16	5859
9	Laminated bed of shale and sandstone	3.8	84	19	377.11	5034
10	Sandstone and intercalation	3.9	79	17	396.45	3534

Assessment of Cavability and Support Resistance Estimation

$$I = \frac{\sigma_c L^2 t^{0.5}}{5}$$

I - caving index

σ_c - compressive strength (kg/cm^2)

L - avg. core length (m)

t - bed thickness (m)

n - constant based on bed RQD

Caving behaviour of different beds of overlying roof depends on bed

- thickness
- strength
- massiveness

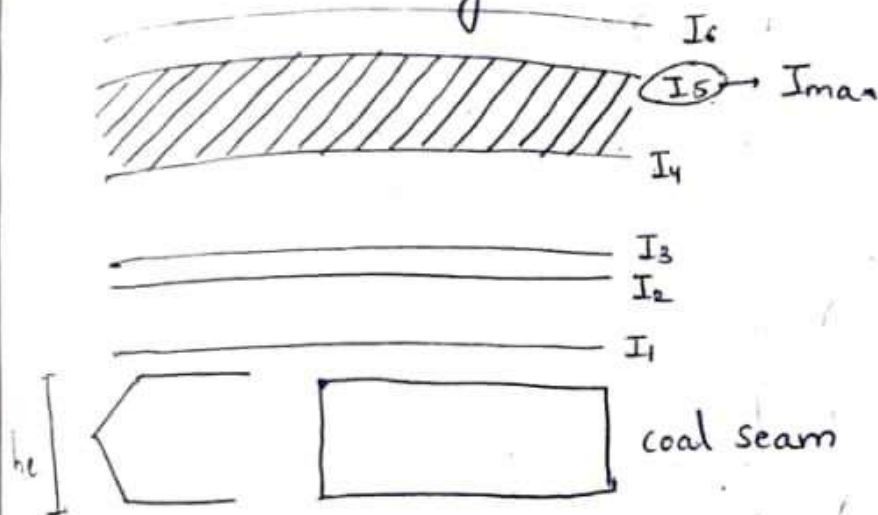
Categorising Cavability of Bed Rock

CATEGORY	NATURE OF CAVING	RANGE OF CAVABILITY INDEX	SPAN
I	Easily Cavable	$I < 2000$	$S_m < 35$
II	Moderately Cavable	$2000 \leq I < 5000$	$35 \leq S_m < 55$
III	Cavable with difficulty	$5000 \leq I < 10000$	$55 \leq S_m < 80$
IV	Cavable with sustainable difficulty	$10000 \leq I < 14000$	$80 \leq S_m < 95$
V	Cavable with extreme difficulty	$I \geq 14000$	$S_m > 95$

Estimation of Support Resistance

Caving properties of the strong bed (I_{max})

Caving index numbers within 12-15 times height based on Bulking Factor



$$I_1 + I_2 + I_3 + I_4$$

I_{max} has potential to support and coal face

layers above I_{max} do participate in loading

$$C_m = \left(\frac{A}{P} + 9.6 h_e \right) + \left(\frac{K I_{max}}{K_1 + 1.5} - 23 \right)$$

A - depends on roof category

C_m - maximum convergence during 1st weight + periodic weights (mm/m)

P - support resistance (t/m²)

I_{max} - caving index of strong bed

K_1 - A factor depending on thickness 't' of caving

bed in blw stony bed + coal seam in terms of extraction height.

he. height of extraction

Roof category I + II

A 1440
1700
1900

III + IV

V

K_1 2 $r \leq 2$
3 $2 < r \leq 4$
5 $4 < r \leq 8$
10 $r \geq 8$

$$K = 0.025$$

Maximum Convergence (Gm)

$$C_m \leq 60 \text{ mm/m}$$

$$60 \text{ mm/m} \leq C_m \leq 100 \text{ mm/m}$$

$$100 \text{ mm/m} \leq C_m \leq 160 \text{ mm/m}$$

$$C_m \geq 160 \text{ mm/m}$$

Expected roof condition

convergence within permissible limit

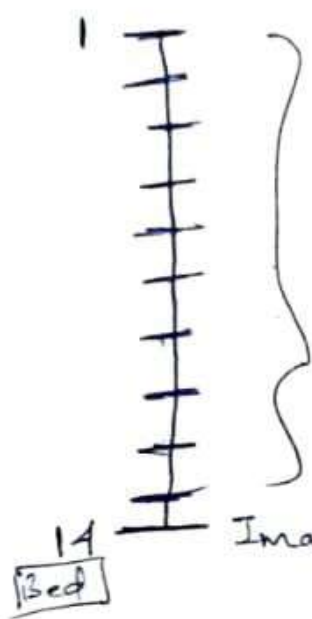
Minor roof fracturing increasing with convergence value
significant roof degradation and roof fracturing. Seriousness ↑ with convergence ↑.
Rock Fall Zone

Maximum convergence $C_m = 60 \text{ mm/m}$

6 a) $H_e = 3.5 \text{ m}$
 $l_c = 4 \text{ m}$
 $W_c = 1.75 \text{ m}$
 $W_d = 0.8 \text{ m}$

$S = ?$
 powered roof support
 capacity

1) $I_{\max}$ Bed 14 $\rightarrow 6197$



2) $1-13 \quad \sum_{i=1}^{13} t = 39.75 \text{ m}$

3) $5000 < I_{\max} < 10000$
 III category

4) $A = 1700$

5) $K = 0.025$

$$C_m = \left(\frac{A}{P} + 9.6 h_e \right) + \left(\frac{K I_{\max}}{K_1 + 1.5} - 23 \right)$$

K_1 nikalne ke liye $\gamma = \frac{t}{h_e} = \frac{39.75}{3.5}$

6) $\gamma = 11.357$

$\gamma > 8$

$K_1 = 10$

7) $C_m = 60$

$$60 = \left[\frac{1700}{P} + 9.6(3.5) \right] + \left[\frac{0.025 \times 6197}{10 + 1.5} - 23 \right]$$

$$60 = \frac{1700}{P} + 33.6 + 13.47 - 23$$

60mm/m se P_e mila uske baad rated P nikalna hai. overall deficiency = 30%

Support Capacity Estimation

$$\text{Support capacity (in tonnes)} = P_r W_c (L_c + W_d + 0.3)$$

$$W_c = 1.75$$

$$L_c = 4.79$$

$$W_d = 0.85$$

P_r - rated support resistance (T/m^2)

W_c - canopy width

L_c - " length

W_d - shearer web depth

Problem 3: $h_e = 3m$

$$G_m = 60 \text{ mm/m}$$

9.36

7414 $G_g F_g$ sst

14.65 tonnes support t

$$\frac{1700}{P} = 36$$

3)

$$P = 47.22 \text{ MPa}$$

support resistance

$$P_{\text{rated}} = 1.3 \times P = 1.3 \times 47.22 = 61.4 \text{ MPa}$$

$$S = P_r \times W_c \times (L_c + W_d + 0.3)$$

$$S = 61.4 \times 1.75 (4 + 0.8 + 0.3)$$

$$S = 548 \text{ Tons}$$

⑥ $H_e' = 4.5 \text{ m}$

$$r = \frac{39.75}{4.5} = 8.83$$

$$r > 8$$

$$K_1 = 10$$